in which *negative cross-side effects* arise. Think about a platform that facilitates the sharing of digital media—music, text, images, videos, and the like. In most circumstances, a growing number of producers (music companies, for example) leads to positive benefits for consumers, but it can also lead to growing complexity and expense—for example, too many varying digital rights management forms to read and accept. When this happens, the cross-side effects flip from positive to negative, leading consumers to abandon the platform or at least reduce their usage. In a similar way, when the proliferation of messages from competing merchants on a platform site leads to unpleasant advertising clutter, the positive impact of expanding producer choice may be transformed into a negative cross-side effect that turns off consumers and damages the platform's value.

We can foresee the arrival of growing pains at Uber as a result of increasing negative cross-side effects. If Uber attracts too many drivers relative to the number of riders, driver downtimes will go up; if Uber attracts too many riders relative to drivers, rider wait times will go up (see Figure 2.2, in which the resulting feedback loops have been inserted).

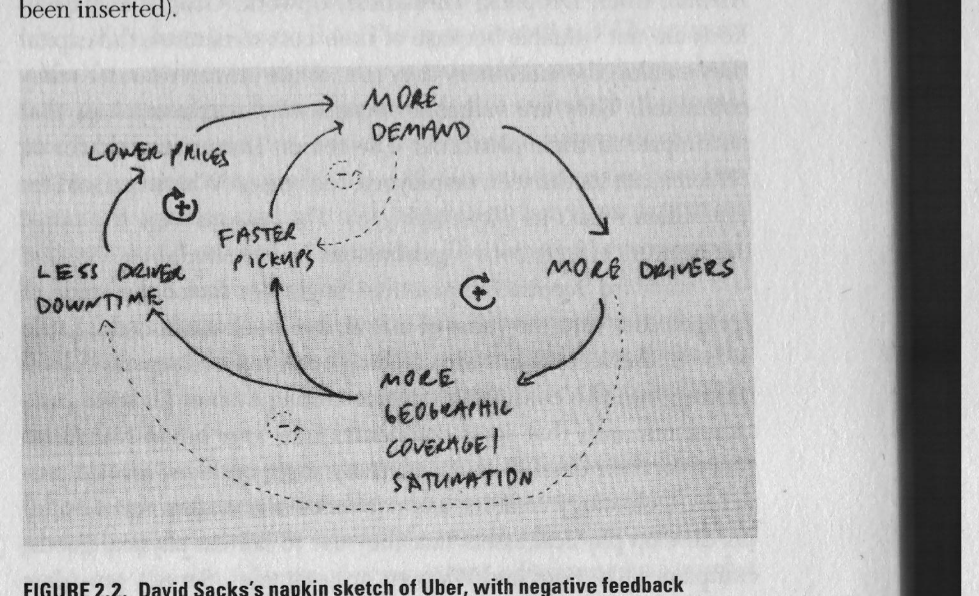

FIGURE 2.2. David Sacks's napkin sketch of Uber, with negative feedback loops inserted.

UnitedHealthcare and Accenture are examples. Technology creators develop and sell forms of intellectual property, such as software and biotechnology; Microsoft and Amgen are examples. And network orchestrators develop networks in which people and companies create value together—in effect, platform businesses. The research suggests that, of the four, network orchestrators are by far the most efficient value creators. On average, they enjoy a market multiplier (based on the relationship between a firm's market valuation and its price-to-earnings ratio) of 8.2, as compared with 4.8 for technology creators, 2.6 for service providers, and 2.0 for asset builders.[16] It's only a slight simplification to say that that quantitative difference represents the value produced by network effects.

Furthermore, where network effects are present, industries operate by different rules.[17] One reason is that it is far easier to scale network effects outside a firm than inside it—since there are always many more people outside a firm than inside it. Thus, where network effects are present, the focus of organizational attention must shift from inside to outside. The firm inverts; it turns inside out. The management of human resources shifts from employees to crowds.[18] Innovation shifts from in-house R & D to open innovation.[19] The primary venue for activities in which value is created for participants shifts from an internal production department to a collection of external producers and consumers—which means that management of *externalities* becomes a key leadership skill. Growth comes not from horizontal integration and vertical integration but from functional integration and network orchestration. The focus on processes such as finance and accounting shifts from cash flows and assets you can own to communities and assets you can influence. And while platform businesses themselves are often extraordinarily profitable, the chief locus of wealth creation is now outside rather than inside the organization.

Network effects are creating the giants of the twenty-first century. Google and Facebook each touch more than one-seventh of the world's population. *In the world of network effects, ecosystems of users are the new source of competitive advantage and market dominance.*

3

ARCHITECTURE

Principles for Designing a Successful Platform

How do we build a platform that invites participation and creates significant value for all its users? How do we provide tools and services that make it easy for producers and consumers to interact in mutually rewarding ways? And how do we design a technological infrastructure that is capable of scaling rapidly, and encouraging positive network effects while minimizing negative ones?

These are daunting challenges. Platforms are complex, multisided systems that must support large networks of users who play different roles and interact in a wide variety of ways. An industry-wide platform—a platform for the health care industry, for example—may need to facilitate interactions among an enormous range of industry participants with motivations that vary widely and change frequently as economic, regulatory, and technological circumstances evolve.

Designers and builders of any complex system often find it difficult to identify a logical starting point. This problem is particularly acute with platform businesses, since they are less familiar and much more complicated than pipeline businesses, which generally feature a straightforward linear design. The natural tendency of those charged with creating a new platform business is to study similar implementations and imitate them. But because no two markets are identical, this strategy often fails. A poorly designed platform produces little or no value for users and generates weak network effects, or none at all.

So where do we start in designing a new platform? The best way

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Exchange of goods or services. As a result of the information exchange, the platform participants may decide to exchange valuable goods or services as well. In some cases, the exchange of goods or services may also occur through the platform. On Facebook, photos, links, and posts with personal or other news are exchanged among users, while on YouTube videos are exchanged. Each item exchanged among platform users can be referred to as a *value unit*. In some cases, platforms feature sophisticated systems that make it easy and convenient for value units to be exchanged. Upwork, for example, provides clients with built-in tools to manage remote service delivery, so digital goods created by a freelancer, like slide decks and videos, can be exchanged directly through the platform itself.

In other cases, goods or services are exchanged outside of the platform (although information about the delivery may be tracked and exchanged on the platform). Transportation services requested via Uber are delivered on real city streets using actual cars; dinner reservations made via Yelp result in physical meals consumed around real tables in actual restaurants.

Exchange of currency. When goods or services are exchanged between platform participants, they are typically paid for using some form of currency. In many cases, this is traditional currency—money transmitted in one of a variety of ways, including credit card data, a PayPal transaction, a Bitcoin transfer, or (rarely) physical cash.

However, there are other forms of value, and therefore other ways in which consumers "pay" producers in the world of platforms. Video viewers on YouTube or followers on Twitter pay the producer with attention, which adds value to the producer in a variety of ways. (If the producer is a political pundit or business leader, for example, he gains value in the form of growing influence as a thought leader; if she is a singer, actor, or athlete, she gains value in the form of a growing fan base.) Community members on sites like TripAdvisor, Dribbble, and 500px pay by enhancing the reputation of producers whose work they like. Thus, attention, fame, influence, reputation, and other intangible forms of value can play the role of "currency" on a platform.

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interaction as easy, attractive, and valuable to users as possible. The fundamental purpose of the platform is to facilitate that core interaction.

This basic rule about the primacy of the core interaction applies despite the fact that many platforms involve a wide range of participants who may interact in a variety of ways. LinkedIn, for example, enables multiple interactions. Professionals exchange ideas about career and business strategy; recruiters exchange information about job listings with potential applicants; human resource managers exchange news about labor market conditions; and thought leaders offer their views about global trends. These various forms of interaction were built into the platform over time, each designed to meet a particular platform goal and to help users create a new form of value. The multisided LinkedIn platform we see today was first designed around a single core interaction: professionals connecting with other professionals.

So let's consider the three key components of the core interaction and how they connect to create value on the platform.

The participants. There are fundamentally two participants in any core interaction: the *producer*, who creates value, and the *consumer*, who consumes value. When defining the core interaction, both roles need to be explicitly described and understood.

One nuance of platform design is recognizing that the same user may play a different role in differing interactions. The same person may be both a host and a guest on Airbnb, although he or she will typically perform only one of those roles in a particular interaction. On YouTube, users may upload videos as well as view them. A well-designed platform makes it easy for users to move from role to role.

Conversely, many users, and many types of users, may perform the same role in an interaction. For example, one of the most common interactions on Facebook is a "status update"—a content posting that informs participants in the network about what a particular member is doing or thinking. The producer who drives a change of status on a particular Facebook page may be an individual, a business, a group of friends, or a nonprofit organization, but the funda-

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In one way or another, every platform makes use of filters to manage the exchange of information. Uber's drivers announce their availability on the platform by sharing various parameters on location, occupancy, and so on—value units that enable them to be matched to the right consumers. When a consumer pulls out her phone and requests a car, she sets up a filter based on her location at the time of the request. Information about the drivers most relevant to the consumer is then exchanged.

Once this exchange of information happens, everything else clicks into action. The car turns up, the traveler is taken to her destination, the appropriate funds are transferred out of the traveler's account, and the driver is compensated. The core interaction is completed—value has been created and exchanged.

Some platforms have more complicated models, but the basic structure remains the same:

Participants + Value Unit + Filter → Core Interaction

Google's search engine acts in a fundamentally similar way. Google's crawlers search the web, creating web page indices (value units). A consumer types in a query. Google combines the query with other specified inputs, such as *social signals*—the volume of "likes," retweets, comments, and other responses received by a particular posting on the Internet. This combination of inputs constitutes the filter, which determines which value units are delivered to the consumer.

On Facebook, your entire network creates status updates, pictures, comments, links, and so on—all of them value units that are being added to the platform. Your news feed algorithm, based on signals you've given in the past through your interactions with previous content, acts as the filter that determines which units are delivered to you and which are not.

When designing a platform, your first and most important job is to decide what your core interaction will be, and then to define the participants, the value units, and the filters to make such core interactions possible.

As we see in cases like LinkedIn and Facebook, platforms often

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capability, became the conduit for the market data the small farmers so desperately needed.

But creating the crucial value units needed to make exchanges possible between the farmers and the *mandis* (the local market-makers) would prove to be an even more significant challenge. "We needed information of various kinds," Choudary explains:

> Of course, we needed price data from the *mandis*—current market rates for various grades of commodities ranging from carrots and cauliflower to beans and tomatoes. This turned out to be fairly easy to gather. Some of the agents themselves would provide the information to us. And we supplemented this source by hiring local people to visit each of the *mandis* to gather price quotes firsthand and report them to us.
>
> The other side of the equation was more difficult. In order to create an electronic information resource that would be truly useful to the farmers, we needed data about the farmers themselves—the crops they were planting, the expected harvest cycle, the locations of the farms, their access to various *mandis*, and so on. All of these factors would influence the best deal they could make in the marketplace.
>
> But gathering this information from a widely scattered collection of farmers—most of them illiterate—was very tricky. We conducted a series of experiments. We tried relying on word of mouth to spread news about the service we were creating and to collect information for us. We tried using the local "head men"—the unofficial mayors of the villages—as conduits for the information. We tried arranging deals with local sellers of seeds, fertilizer, and SIM cards for cell phones, all of whom were frequently in touch with individual farmers. But none of these methods worked well—the people we tried to work with weren't interested, and the incentives simply weren't strong enough to produce a powerful flow of data.
>
> In the end, we had to build our own network of data gatherers—what Indians call a "feet on street" (FOS) sales force. The FOS team went door to door in each village, meeting with farmers and recording key information about their crops and their marketing plans on paper forms. Then they brought the data back to our offices, where we entered

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match participants accurately will waste their time and energy, soon driving participants away.

Let's examine each of these three crucial functions in a little more detail. Effective platform design is all about creating systems that perform these functions as powerfully as possible.

Pull. Attracting consumers to platforms presents challenges that pipeline companies don't face. Consequently, the approach to marketing such platforms is likely to seem counterintuitive, especially to business leaders who grew up in the old pipeline-dominated universe.

To begin with, platforms need to solve a chicken-or-egg problem that pipeline businesses don't suffer from: users won't come to a platform unless it has value, and a platform won't have value unless users use it. Most platforms fail simply because they never overcome this problem. It's such an important challenge that we'll devote all of chapter 5 to analyzing it and helping you solve it.

A second pull challenge revolves around keeping the interest of users who visit or sign up for the platform. The large social networks of our day have all faced this problem at some point. Facebook, for example, discovered that users found the platform valuable only after they had connected to a minimum number of other users. Until then, they were likely to stop using the network entirely. In response, Facebook shifted its marketing efforts away from recruiting new members to helping them form connections.

One powerful tool that encourages users to keep returning to the platform is the *feedback loop*. A feedback loop in a platform may take various forms, all of which serve to create a constant stream of self-reinforcing activity. In the typical feedback loop, a flow of value units is that generates a response from the user. If the units are relevant and interesting, the user will be drawn to the platform repeatedly, generating a further flow of value units and facilitating more interactions. Effective feedback loops help to swell the network, increase value creation, and enhance network effects.

One kind of feedback loop is the *single-user feedback loop*. This involves an algorithm built into the platform infrastructure that ana-

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value can be created and exchanged, and lay out principles that govern these interactions. That's what the process of facilitating is all about.

One aspect of facilitating interactions is making it as easy as possible for producers to create and exchange valuable goods and services via the platform. This may involve providing creative tools for collaboration and sharing, as the Canadian photography platform 500px does with its infrastructure, which allows photographers to host their entire portfolios on the platform, or as invention platform Quirky does with its tools for letting users work together on creative ideas for innovative products and services.

Facilitating interactions may also involve reducing barriers to usage. Not long ago, a Facebook user who wanted to share photos with friends had to use a camera, transfer the images to a computer, use Photoshop or another software package to edit them, and finally upload them to Facebook. Instagram enabled users to snap, modify, and share pictures in three clicks on a single device. Lowering barriers to usage in this way encourages interactions and helps expand participation on the platform.

In some cases, increasing barriers has a positive effect on usage. Sittercity is platform that helps parents find babysitters. To inspire trust among its users (the parents), Sittercity has imposed a stringent set of rules that restrict those who can sign up as producers (the babysitters). In other cases, platforms must develop intrusive rules for curating value units and other producer-created content in order to encourage desirable interactions and discourage undesirable ones. Though relatively rare, misdeeds like the racist and sexist abuse spewed by trolls on Reddit, the murders of people found through Craigslist, and the trashing of apartments booked through Airbnb illustrate how undesirable interactions damage network effects.

Designing a platform to facilitate value-creating interactions is not a simple matter. We'll explore the challenges of platform curation and governance in more detail in chapters 7 and 8.

Match. A successful platform creates efficiencies by matching the right users with one another and ensuring that the most relevant goods